

JOTIRAM KALE

Generative AI & Agentic AI Application Developer

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Summary

Generative AI developer and final-year Computer Science student with hands-on experience building end-to-end LLM and agentic AI applications using Python, Flask, Streamlit, LangChain, and LangGraph. Skilled in Retrieval-Augmented Generation (RAG), prompt engineering, vector databases (ChromaDB), human-in-the-loop agent workflows, anomaly detection, and forecasting. Deployed projects in document Q&A and personal finance analytics using Groq LLM, Sentence-Transformers, and OCR. Passionate about building production-ready AI systems that solve real-world problems.

Skills

Programming Languages: Python

Generative AI & Agentic AI: LangChain, LangGraph, LLM's, RAG, Prompt Engineering, Human-in-the-Loop, LLM Applications, Document Q&A, Groq API

Vector Databases: ChromaDB

Frameworks & Libraries: Flask, Streamlit, Sentence-Transformers,

Tools: Git, GitHub, Jupyter Notebook, REST APIs

Projects

Personal Finance Agent | *Python, LangGraph, LangChain, Pandas, Streamlit, Groq LLM*

Sept 2026

- Built an agentic AI-powered personal finance system that analyzes bank and credit-card transaction data, categorizes expenses, detects anomalies, forecasts future spending, and generates actionable financial recommendations.
- Designed a LangGraph-based workflow with human-in-the-loop review and approval checkpoints, enabling safe, user-controlled decision making before executing recommended actions.
- Developed anomaly detection pipelines to identify duplicate transactions, unusual spending patterns, new merchants, and recurring subscription price changes using transaction history analysis.
- Implemented spending forecast models using historical transaction trends to estimate category-wise monthly expenses and support budget planning.
- Created a rule-based merchant categorization engine that learns user-defined mappings, improving classification consistency and reducing repeated manual review.
- Developed an interactive Streamlit dashboard for transaction analytics, anomaly reporting, spending forecasts, and approval-based action management.

AI PDF Assistant | *Python, Flask, LangChain, ChromaDB, Groq LLM*

Aug 2026

- Developed a Retrieval-Augmented Generation (RAG) pipeline that ingests PDFs, chunks text with 200-token overlap, and stores embeddings in ChromaDB for semantic search.
- Integrated Sentence-Transformers (`all-MiniLM-L6-v2`) for document embeddings and Groq-hosted LLM via LangChain for grounded answer generation.
- Engineered a hallucination-safe prompt that restricts LLM answers to retrieved context only, returning a fallback message when information is missing.
- Architected a modular **LangChain LCEL** pipeline (`prompt | llm | StrOutputParser`) to decouple prompt logic, LLM selection, and output formatting, enabling seamless swapping of underlying models.
- Built a Flask web interface with file upload and natural-language Q&A, enabling conversational document analysis in real time.
- GitHub: github.com/jotiramkale/pdfassistant

Education

B.Tech. Computer Science and Engineering

D.Y. Patil Technical Campus, Talsande-Kolhapur

Expected: 2027

CGPA: 6.9 / 10

Diploma in Computer Technology

Solapur Education Society's Polytechnic, Solapur

Completed: 2024

Percentage: 83%